Xiang MENG

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Academic Positions

Post-Doctoral Researcher, Dana-Farber Cancer Institute, Data Science *2025 – Present*

Post-Doctoral Research Affiliate, Harvard T.H. Chan School of Public Health, Biostatistics *2025 – Present*

Education

PhD, Harvard University, Statistics *2025*

Advisors: Luke Miratrix, Iavor Bojinov, Natesh Pillai

MSc, University of Washington (UW), Statistics *2020*

BSc, National University of Singapore (NUS), Quantitative Finance and Statistics *2018*

Minor in Computer Science. Exchange program in University of California, San Diego (UCSD).

Research Interests

Causal Inference and Survival Analysis for Cancer Clinical Trials; External Controls and Surrogate Endpoints; Causal and Generative AI for Clinical Data; Matching Methods; Semiparametric Theory and Debiased Machine Learning

Selected Achievements & Awards

- Selected as Dana-Farber's institutional nominee, through internal competition, for the NCI K99/R00 Pathway to Independence Award (application submitted Oct 2026) 2026
- Rose Fellowship, Department of Epidemiology, Harvard T.H. Chan School of Public Health ($5,000 extended travel award) 2026
- Rising Stars in Data Science, selected participant recognized for innovative research and career potential, hosted by UC San Diego, University of Chicago, and Stanford University 2024
- Bok Teaching Certificate, Harvard University 2024
- Lijen Industrial Development Medal for best thesis project in the discipline, NUS 2018

Publications

† Student I supervised or co-supervised. * Equal contribution.

Peer-reviewed publications

[1] **Meng, X.***, Che, J.*, Miratrix, L. (2026, in press). Caliper Synthetic Matching: Generalized Radius Matching with Local Synthetic Controls. *Political Analysis*. *arXiv:2411.05246.*

[2] **Meng, X.**, Smith, A. (2026). ScoreCausal: Learning the causal embedding directly via score-ratio dimension reduction. *NeurIPS 2026 Workshop on World Models for High-Stakes Health (WMHS)*; also accepted at STAI-X.

[3] **Meng, X.**, Huang, J. (2025). REFINE2: A simplified simulation tool to help epidemiologists evaluate the suitability and sensitivity of effect estimation within user-specified data. *American Journal of Epidemiology*. https://doi.org/10.1093/aje/kwaf195

[4] Weingarden, H., **Meng, X.**, Armey, M., Onnela, J.-P., Jaroszewski, A., Armstrong, C., Wilhelm, S. (2025). Predicting the strength of next-day negative emotion states in body dysmorphic disorder using passive smartphone data: An intensive longitudinal assessment study. *Internet Interventions*, 100833.

[5] Yang, L., Raveendran, G., **Meng, X.**, Lin, J., Meng, Z. (2024). Predict progression-free survival and overall survival using objective response rate for anti-PD1/PDL1 therapy development. *BMC Cancer*, 24, 912. https://doi.org/10.1186/s12885-024-12664-1

[6] Wang, L., **Meng, X.**, Richardson, T., Robins, J. (2022). Coherent modeling of longitudinal causal effects on binary outcomes. *Biometrics*. doi:10.1111/biom.13687

[7] Gordon, E., **Meng, X.**, Barnes, M., Bhattacharjee, T., Srinivasa, S. (2020). Adaptive robot-assisted feeding: An online learning framework for acquiring previously unseen food items. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Manuscripts under review and preprints

[1] **Meng, X.**, Bojinov, I. Time-varying causal survival learning. *arXiv:2503.00730*. Preliminary version at AAAI-25 Workshop on Artificial Intelligence with Causal Techniques; journal version in preparation.

[2] Nabi, K.†, **Meng, X.**, Tian, L., Connors, J., Schrag, D., Uno, H. Average cause-specific hazard: A censoring-invariant measure of event burden under competing risks. *Major revision, Biometrics*.

[3] **Meng, X.**, Tian, L., Kehl, K., Uno, H. Efficient and debiased learning of average hazards under non-proportional hazards. *Under review, Statistical Methods in Medical Research*. *arXiv:2602.13475.*

[4] **Meng, X.**, Smith, A., Miratrix, L. Simple and reliable inference for matching estimators: A single-matching variance approach. *Under review, Journal of Econometrics*. *arXiv:2506.11317.*

[5] **Meng, X.**, Tian, L., Uno, H. A cosine-projection framework for choosing among tests based on the Cox model, restricted mean survival time, and average hazard in survival trials. *Under review, Biometrics*. *arXiv:2610.15149.*

[6] **Meng, X.**, Dempsey, W., Liao, P., Reid, N., Klasnja, P., Murphy, S. Evaluation of the HeartSteps online sampling algorithm. *Revision in preparation*. *arXiv:2501.02137.*

Working papers

[1] Kehl, K., **Meng, X.** Oncology causal inference: Learning from past decisions to inform future care.

[2] **Meng, X.** Causal matching with a rank-learned prognostic score.

[3] **Meng, X.**, Wei, Y.†, Uno, H. Generalized method of moments estimation for average hazard regression using pseudo-observations.

[4] Hong, X.†, **Meng, X.**, Uno, H. Inverse probability of treatment weighted average hazard with survival weight for censored time-to-event outcomes.

[5] **Meng, X.**, Han, F. Bias ignorability of higher-order kernel matching estimators in causal inference.

Software & Methodologies

Xiang Meng, Jonathan Che, Luke Miratrix. Caliper Synthetic Matching (CSMatch). R package. Available at https://github.com/MiratrixCARESLab/CSMatch.

Xiang Meng, Jonathan Che, Luke Miratrix. Replication Data for: "Caliper Synthetic Matching: Generalized Radius Matching with Local Synthetic Controls". *Political Analysis Dataverse, Harvard Dataverse,* doi:10.7910/DVN/5HB8LO

Linbo Wang, **Xiang Meng**, Thomas Richardson, James Robins. Methodology of "Coherent modeling of longitudinal causal effects on binary outcomes". *Harvard Dataverse,* https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/UZAI96&faces-redirect=true

Xiang Meng and Jonathan Huang. "REFINE2: A tool to evaluate real-world performance of machine-learning based effect estimators for molecular and clinical studies". *Github,* https://github.com/mengeks/drml-plasmode

Invited Talks, Contributed Conference and Poster Presentations

"A Cosine-Projection Framework for Choosing Among Cox HR, D-RMST, and Average-Hazard Tests in Survival Trials"

CompBio Seminar, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — *Abu Dhabi, UAE, Jul 2026*
International Biometric Conference (IBC) — *Seoul, South Korea, Jul 2026*
Tokyo Medical University — *Tokyo, Japan, Jul 2026*

"Efficient and Debiased Learning of Average Hazards Under Non-Proportional Hazards"

Joint Statistical Meetings (JSM) — *Boston, MA, Aug 2026*
American Causal Inference Conference (ACIC) — *Salt Lake City, UT, May 2026*

"Simple and Reliable Inference for Matching Estimators" (Past title: "Variance estimation after matching or re-weighting")

Seoul National University (SNU) Statistics Seminar — *Seoul, South Korea, Jul 2026*
Harvard Applied Statistics Seminar — *Cambridge, MA, Apr 2026*
American Causal Inference Conference (ACIC) — *Detroit, MI, May 2025*

"Time-Varying Causal Survival Learning"

American Causal Inference Conference (ACIC) — *Detroit, MI, May 2025*
AAAI-25 AI for Causal Techniques — *Philadelphia, PA, Mar 2025*

Harvard Dept. Government Applied Statistics Workshop — *Cambridge, MA, Nov 2024*
Rising Star Workshop, UCSD — *San Diego, CA, Nov 2024*

"Caliper Synthetic Matching"

Peking University School of Education, invited talk — *Beijing, China, Jan 2025*
New England Statistics Symposium (NESS) — *Storrs, CT, May 2024*
American Causal Inference Conference (ACIC) — *Seattle, WA, May 2024*
New England Student Research Symposium on Statistics and Data Science — *Boston, MA, Apr 2024*

"A Novel Evaluation of a Just-in-Time Adaptive Intervention Algorithm"

INFORMS Annual Meeting — *Toronto, ON, Oct 2023*
Eastern North American Region (ENAR) Meeting — *Online, Mar 2022*
14th CMStatistics, Advances in statistical methods for mobile health — *Online, Dec 2021*
3rd Harvard Health Data Science Symposium — *Boston, MA, Nov 2021*
Joint Statistical Meetings (JSM) — *Online, Aug 2021*

"Causal Questions in Micro-Randomized Trials (MRTs): Introduction and Challenges"

Society for Causal Inference (SCI) Causal Inference for Social Impact — *Online, Jun 2021*

"Learning from failures in robot-assisted feeding"

IJCAI 2019 Workshop on AI× Food, Poster and Oral Presentation — *Macao, China, Aug 2019*

Teaching Experience

Introduction to Bioinformatics, Harvard University, Head Teaching Fellow (TF). Designed weekly section materials. — *Spring 2025*

Introduction to Data Science, Harvard University, Head Teaching Fellow (TF) for 220 students, oversaw 29 TFs. — *Fall 2023*

The Art of Teaching in Statistics, Harvard University, teaching preparation for all 1st year statistics PhD students. — *AY 2022 – 23*

Data Analysis in Modern Biostatistics (Biomarker and Cancer Research), Harvard University — *Spring 2022*

Introduction to Probability, Harvard University — *Fall 2021*

STEMPREP Summer Statistics Course for 7th and 8th Grade Students, UW — *Summer 2019*

Programming Methodology in Python, NUS — *AY 2015–16, Spring 2018*

Research Mentoring

Khondoker Nabi, PhD student in Biostatistics, Harvard University. Co-supervised with H. Uno. — *2025 – Present*
Average cause-specific hazard under competing risks; first-author paper in major revision at Biometrics.

Xiong Hong, PhD student in Biostatistics, University of Pennsylvania. Co-supervised with H. Uno. — *2025 – Present*
Inverse-probability-weighted average hazard for observational time-to-event data; first-author manuscript in preparation.

Yuze Wei, Master's student in Data Science, Harvard University. Supervised. — *2025 – Present*
Average-hazard regression via pseudo-observations; randomization inference for survival outcomes with few treated patients.

Academic & Student Service

Reviewer for Biometrika, Journal of Clinical Epidemiology (JCE), Lifetime Data Analysis (LDA), Biometrical Journal, Journal of Applied Statistics (JAS), American Causal Inference Conference (ACIC), NeurIPS workshop.

Co-Chair for seminar department at Harvard Chinese Student & Scholar Association (HCSSA), organized and supervised 7 seminars to give industry and academia exposure to Chinese graduate students across art and science, industry and academia.

Volunteer service at more than 15 events supporting older adults, people with disabilities, women, and gender-diverse individuals.

Industry Experience

Sanofi *May – Aug 2022*

Summer Intern, Clinical Evidence and Data Science

- Co-created a comprehensive database from over 400 clinical trials on Anti-PD1/PDL1 cancer treatments using R.
- Co-developed interpretable models relating objective response rate to progression-free and overall survival.

Skills

Languages: Mandarin Chinese (Native), English and Japanese (Proficient in writing, speaking, reading, listening)
Programming: R, Python, SQL, SAS, C++, Unix, Matlab

References

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